Quotient Rings, Spectra, and Cyclic Codes

From $\mathbb{Z}$ and $\mathbb{C}[x]$ to $\mathbb{F}_q[x]/\langle x^n - 1\rangle$

Contents

1 Introduction

These notes have three goals.

(1) To understand the ideal correspondence between a ring R and its quotient

$$R \longrightarrow R/I.$$

(2) To interpret this correspondence geometrically through spectra, first in the familiar examples

$$\mathbb{Z} \qquad \text{and} \qquad \mathbb{C}[x].$$

(3) To apply the same idea to the quotient ring

$$\mathbb{F}_q[x]/\langle x^n - 1\rangle$$

and explain why its ideals are exactly the cyclic codes of length n over $\mathbb{F}_q$.

The central philosophy is the following:

> Passing from R to R/I means forcing the elements of I to become zero. Therefore only those ideals of R that already contain I survive in the quotient.

In the cyclic-code setting, the additional feature is that the quotient ring

$$\mathbb{F}_q[x]/\langle x^n - 1\rangle$$

is naturally identified with the vector space $\mathbb{F}_q^n$, and multiplication by x becomes cyclic shift of coordinates. This turns ring-theoretic ideals into coding-theoretic cyclic codes.

2 The ideal correspondence for a quotient ring

We begin with the basic algebraic fact.

Theorem 2.1 (Ideal correspondence)**.** *Let R be a ring, let $I \lhd R$ be an ideal, and let*

$$\pi : R \longrightarrow R/I$$

be the quotient map. Then the assignment

$$J' \longmapsto \pi^{-1}(J')$$

defines an inclusion-preserving bijection

$$\{\textit{ideals of } R/I\} \longleftrightarrow \{\textit{ideals } J \lhd R \textit{ such that } I \subseteq J\}.$$

Its inverse is given by

$$J \longmapsto J/I.$$

Proof. Let $J' \lhd R/I$. Then $\pi^{-1}(J')$ is an ideal of R. Since $\pi(I) = 0$, we have

$$I \subseteq \pi^{-1}(J').$$

Conversely, let $J \lhd R$ with $I \subseteq J$. Then

$$J/I = \{\overline{x} \in R/I : x \in J\}$$

is an ideal of R/I.

Now we check that these constructions are inverse to one another.

If $J \lhd R$ with $I \subseteq J$, then

$$\pi^{-1}(J/I) = J.$$

Indeed, $\pi(x) \in J/I$ if and only if $x \in J$.

If $J' \lhd R/I$, then

$$\pi^{-1}(J')/I = J'.$$

Indeed, every element of J' has the form $\overline{x}$ with $x \in \pi^{-1}(J')$.

Thus the correspondence is bijective, and it is plainly inclusion-preserving. $\square$

Remark 2.2. This theorem is often summarized by the slogan

$$\text{ideals of } R/I \quad \leftrightarrow \quad \text{ideals of } R \text{ containing } I.$$

3 First model: the quotient $\mathbb{Z} \to \mathbb{Z}/12\mathbb{Z}$

The ring $\mathbb{Z}$ is the simplest principal ideal domain. Every ideal has the form

$$(m) = m\mathbb{Z}$$

for some integer $m \geq 0$.

Now consider the quotient map

$$\pi : \mathbb{Z} \longrightarrow \mathbb{Z}/12\mathbb{Z}.$$

Proposition 3.1. *The ideals of $\mathbb{Z}/12\mathbb{Z}$ correspond exactly to the ideals $(m) \subseteq \mathbb{Z}$ such that*

$$(12) \subseteq (m).$$

Equivalently, they correspond exactly to the positive divisors of 12.

Proof. By Theorem 2.1, the ideals of $\mathbb{Z}/12\mathbb{Z}$ correspond to the ideals of $\mathbb{Z}$ containing (12). Since every ideal of $\mathbb{Z}$ is of the form (m), this means

$$(12) \subseteq (m).$$

This inclusion holds if and only if m divides 12. $\square$

Thus the ideals of $\mathbb{Z}/12\mathbb{Z}$ correspond to

$$1,\ 2,\ 3,\ 4,\ 6,\ 12.$$

More explicitly, the corresponding ideals downstairs are

$$(1)/(12) = \mathbb{Z}/12\mathbb{Z},$$

$$(2)/(12), \quad (3)/(12), \quad (4)/(12), \quad (6)/(12),$$

and

$$(12)/(12) = 0.$$

Example 3.2. The ideal $(4)/(12)$ consists of the residue classes

$$\{\,\overline{0}, \overline{4}, \overline{8}\,\} \subseteq \mathbb{Z}/12\mathbb{Z}.$$

It is the image of the upstairs ideal $(4) \subseteq \mathbb{Z}$.

The moral is clear:

> Passing from $\mathbb{Z}$ to $\mathbb{Z}/12\mathbb{Z}$ means forcing $12 = 0$. Therefore only those ideals of $\mathbb{Z}$ compatible with the relation $12 = 0$ remain visible.

4 Second model: the quotient $\mathbb{C}[x] \to \mathbb{C}[x]/\langle g(x)\rangle$

The polynomial ring $\mathbb{C}[x]$ is also a principal ideal domain, so every ideal has the form

$$(f(x))$$

for some polynomial $f(x) \in \mathbb{C}[x]$.

Let

$$\pi : \mathbb{C}[x] \longrightarrow \mathbb{C}[x]/\langle g(x)\rangle$$

be the quotient map.

Proposition 4.1. *The ideals of $\mathbb{C}[x]/\langle g(x)\rangle$ correspond exactly to the ideals $(f(x)) \subseteq \mathbb{C}[x]$ satisfying*

$$(g(x)) \subseteq (f(x)).$$

Equivalently, they correspond exactly to the divisors of $g(x)$, up to nonzero scalar multiples.

Proof. By Theorem 2.1, the ideals of the quotient correspond to ideals of $\mathbb{C}[x]$ containing $(g(x))$. Since $\mathbb{C}[x]$ is a PID, those ideals are exactly the ideals $(f(x))$ such that

$$(g(x)) \subseteq (f(x)).$$

In a PID, the inclusion

$$(g) \subseteq (f)$$

holds if and only if f divides g. $\square$

Example 4.2. Let

$$g(x) = x^2(x-1).$$

Then the monic divisors of $g(x)$ are

$$1, \quad x, \quad x^2, \quad x-1, \quad x(x-1), \quad x^2(x-1).$$

Hence the ideals of

$$\mathbb{C}[x]/\langle x^2(x-1)\rangle$$

are precisely

$$(1)/(g), \quad (x)/(g), \quad (x^2)/(g), \quad (x-1)/(g), \quad (x(x-1))/(g), \quad (g)/(g).$$

Thus the polynomial case is perfectly parallel to the integer case:

$$\begin{aligned} \text{divisors of } 12 \quad &\longleftrightarrow \quad \text{ideals of } \mathbb{Z}/12\mathbb{Z}, \\ \text{divisors of } g(x) \quad &\longleftrightarrow \quad \text{ideals of } \mathbb{C}[x]/\langle g(x)\rangle. \end{aligned}$$

5 Spectra, points, and residue fields

We now reinterpret the quotient construction geometrically.

5.1 Closed subsets cut out by ideals

Let R be a ring and $I \triangleleft R$. The quotient map

$$R \longrightarrow R/I$$

induces a map of spectra

$$\mathrm{Spec}(R/I) \longrightarrow \mathrm{Spec}(R).$$

Its image is the closed subset

$$V(I) = \{\mathfrak{p} \in \mathrm{Spec}(R) : I \subseteq \mathfrak{p}\}.$$

Hence there is a canonical homeomorphism

$$\mathrm{Spec}(R/I) \cong V(I)$$

where $V(I)$ carries the subspace topology from $\mathrm{Spec}(R)$.

Thus quotienting by I corresponds geometrically to restricting to the closed subspace cut out by the equation $I = 0$.

5.2 Points and residue fields

For a point $x \in \mathrm{Spec}(R)$, represented by a prime ideal $\mathfrak{p}_x$, the residue field is

$$\kappa(x) = \mathrm{Frac}(R_{\mathfrak{p}_x}/\mathfrak{p}_x R_{\mathfrak{p}_x}).$$

Equivalently, if $\mathfrak{p}_x$ is maximal, then simply

$$\kappa(x) \cong R/\mathfrak{p}_x.$$

There is a canonical map

$$\mathrm{Spec}(\kappa(x)) \longrightarrow \mathrm{Spec}(R)$$

whose image is the point x.

Remark 5.1. It is important not to confuse the quotient map

$$R \to R/I$$

with the family of point maps

$$\mathrm{Spec}(\kappa(x)) \to \mathrm{Spec}(R).$$

The quotient map cuts out a closed subset. The residue-field maps describe the individual points of the space.

6 The spectrum of $\mathbb{Z}$

The points of $\mathrm{Spec}(\mathbb{Z})$ are:

- the generic point (0);
- the closed points (p) for prime integers p.

At a closed point (p), the residue field is

$$\kappa((p)) \cong \mathbb{Z}_{(p)}/p\mathbb{Z}_{(p)} \cong \mathbb{F}_p.$$

So each closed point gives a map

$$\mathrm{Spec}(\mathbb{F}_p) \longrightarrow \mathrm{Spec}(\mathbb{Z}).$$

At the generic point (0), the residue field is

$$\kappa((0)) = \mathbb{Q}.$$

So there is also a map

$$\mathrm{Spec}(\mathbb{Q}) \longrightarrow \mathrm{Spec}(\mathbb{Z}).$$

Remark 6.1. The family

$$\coprod_{p \text{ prime}} \mathrm{Spec}(\mathbb{F}_p) \longrightarrow \mathrm{Spec}(\mathbb{Z})$$

captures all closed points, but it does *not* include the generic point. To include every point, one should write

$$\coprod_{x \in \mathrm{Spec}(\mathbb{Z})} \mathrm{Spec}(\kappa(x)) \longrightarrow \mathrm{Spec}(\mathbb{Z}).$$

6.1 A quotient example

Consider the quotient

$$\mathbb{Z} \longrightarrow \mathbb{Z}/12\mathbb{Z}.$$

Then

$$\mathrm{Spec}(\mathbb{Z}/12\mathbb{Z}) \cong V(12) \subseteq \mathrm{Spec}(\mathbb{Z}).$$

Set-theoretically,

$$V(12) = \{(2), (3)\},$$

because the prime ideals containing (12) are exactly (2) and (3).

Thus quotienting by (12) removes everything except the closed points lying over the prime divisors of 12.

7 The spectrum of $\mathbb{C}[x]$

The prime ideals of $\mathbb{C}[x]$ are:

- the generic point (0);
- the closed points $(x - \alpha)$ for $\alpha \in \mathbb{C}$.

At a closed point $(x - \alpha)$, the residue field is

$$\kappa((x - \alpha)) \cong \mathbb{C}[x]/(x - \alpha) \cong \mathbb{C}.$$

So each closed point gives a map

$$\mathrm{Spec}(\mathbb{C}) \longrightarrow \mathrm{Spec}(\mathbb{C}[x]).$$

At the generic point (0), the residue field is

$$\kappa((0)) = \mathbb{C}(x).$$

Remark 7.1. The family

$$\coprod_{\alpha \in \mathbb{C}} \mathrm{Spec}(\mathbb{C}) \longrightarrow \mathrm{Spec}(\mathbb{C}[x])$$

captures all closed points, but omits the generic point. The complete pointwise description is

$$\coprod_{x \in \mathrm{Spec}(\mathbb{C}[x])} \mathrm{Spec}(\kappa(x)) \longrightarrow \mathrm{Spec}(\mathbb{C}[x]).$$

7.1 A quotient example

Let $g(x) \in \mathbb{C}[x]$ and consider

$$\mathbb{C}[x] \longrightarrow \mathbb{C}[x]/\langle g(x) \rangle.$$

Then

$$\mathrm{Spec}(\mathbb{C}[x]/\langle g(x) \rangle) \cong V(g) \subseteq \mathrm{Spec}(\mathbb{C}[x]).$$

If

$$g(x) = \prod_{i=1}^{r} (x - \alpha_i)^{e_i},$$

then, set-theoretically,

$$V(g) = \{(x - \alpha_1), \ldots, (x - \alpha_r)\}.$$

Thus quotienting by $\langle g(x) \rangle$ cuts out the roots of $g(x)$ as points of the affine line.

Proposition 7.2 (Squarefree case and Chinese remainder theorem)**.** *Suppose*

$$g(x) = \prod_{i=1}^{r} (x - \alpha_i)$$

has distinct roots. Then

$$\mathbb{C}[x]/\langle g(x) \rangle \cong \prod_{i=1}^{r} \mathbb{C}[x]/\langle x - \alpha_i \rangle \cong \prod_{i=1}^{r} \mathbb{C}.$$

Consequently,

$$\mathrm{Spec}(\mathbb{C}[x]/\langle g(x) \rangle) \cong \coprod_{i=1}^{r} \mathrm{Spec}(\mathbb{C}).$$

Proof. Since the ideals $(x - \alpha_i)$ are pairwise comaximal, the Chinese remainder theorem gives

$$\mathbb{C}[x]/\langle \prod_{i=1}^{r} (x - \alpha_i) \rangle \cong \prod_{i=1}^{r} \mathbb{C}[x]/\langle x - \alpha_i \rangle.$$

Each factor is canonically isomorphic to $\mathbb{C}$. $\square$

Remark 7.3. If $g(x)$ has repeated factors, then the underlying topological space still consists of the roots of g, but the quotient ring remembers the multiplicities through nilpotent elements.

8 The cyclic-code quotient ring

We now turn to coding theory.

Fix a finite field $\mathbb{F}_q$ and a positive integer n. Define

$$A := \mathbb{F}_q[x]/\langle x^n - 1\rangle.$$

By the ideal correspondence, the ideals of A are exactly the ideals of $\mathbb{F}_q[x]$ containing $\langle x^n - 1\rangle$. Since $\mathbb{F}_q[x]$ is a PID, every such ideal has the form $\langle g(x)\rangle$ where

$$g(x) \mid x^n - 1.$$

Hence:

Proposition 8.1. *The ideals of $\mathbb{F}_q[x]/\langle x^n - 1\rangle$ are exactly the ideals*

$$\langle\overline{g(x)}\rangle \subseteq \mathbb{F}_q[x]/\langle x^n - 1\rangle,$$

where $g(x)$ runs through the monic divisors of $x^n - 1$.

This is formally identical to the earlier examples:

$$\begin{gathered}\text{ideals of } \mathbb{Z}/12\mathbb{Z} \leftrightarrow \text{divisors of } 12,\\ \text{ideals of } \mathbb{C}[x]/\langle g(x)\rangle \leftrightarrow \text{divisors of } g(x),\\ \text{ideals of } \mathbb{F}_q[x]/\langle x^n - 1\rangle \leftrightarrow \text{divisors of } x^n - 1.\end{gathered}$$

9 From the quotient ring to the vector space $\mathbb{F}_q^n$

Every class in A has a unique representative of degree $< n$, namely

$$a_0 + a_1x + \cdots + a_{n-1}x^{n-1}.$$

Hence there is a natural $\mathbb{F}_q$-linear isomorphism

$$\Phi : A \longrightarrow \mathbb{F}_q^n, \qquad \Phi\Big(\overline{a_0 + a_1x + \cdots + a_{n-1}x^{n-1}}\Big) = (a_0, a_1, \ldots, a_{n-1}).$$

Thus, as an $\mathbb{F}_q$-vector space,

$$A \cong \mathbb{F}_q^n.$$

9.1 Multiplication by x becomes cyclic shift

Consider the element

$$f(x) = a_0 + a_1x + \cdots + a_{n-1}x^{n-1}.$$

Then in A,

$$xf(x) = a_0x + a_1x^2 + \cdots + a_{n-2}x^{n-1} + a_{n-1}x^n.$$

Since

$$x^n \equiv 1 \pmod{x^n - 1},$$

we obtain

$$xf(x) \equiv a_{n-1} + a_0x + a_1x^2 + \cdots + a_{n-2}x^{n-1}.$$

Therefore under Φ, multiplication by x corresponds to the cyclic shift

$$(a_0, a_1, \ldots, a_{n-1}) \longmapsto (a_{n-1}, a_0, \ldots, a_{n-2}).$$

This is the fundamental bridge between algebra and coding theory.

10 Cyclic codes as ideals

We now give the main theorem.

Definition 10.1. A *cyclic code* of length n over $\mathbb{F}_q$ is an $\mathbb{F}_q$-linear subspace

$$C \subseteq \mathbb{F}_q^n$$

such that whenever

$$(c_0, c_1, \ldots, c_{n-1}) \in C,$$

the cyclic shift

$$(c_{n-1}, c_0, \ldots, c_{n-2})$$

also belongs to C.

Theorem 10.2 (Cyclic codes are ideals)**.** *Under the identification*

$$\Phi : \mathbb{F}_q[x]/\langle x^n - 1\rangle \xrightarrow{\sim} \mathbb{F}_q^n,$$

the cyclic codes of length n over $\mathbb{F}_q$ are exactly the ideals of

$$\mathbb{F}_q[x]/\langle x^n - 1\rangle.$$

Proof. Let $C \subseteq \mathbb{F}_q^n$ be a cyclic code. Since C is an $\mathbb{F}_q$-subspace, the set

$$\Phi^{-1}(C) \subseteq A$$

is closed under addition and scalar multiplication. Since C is closed under cyclic shift and multiplication by x corresponds to cyclic shift, $\Phi^{-1}(C)$ is closed under multiplication by x.

Now let

$$a(x) = a_0 + a_1 x + \cdots + a_m x^m \in \mathbb{F}_q[x]$$

and let $c \in \Phi^{-1}(C)$. Then

$$a(x)c = a_0 c + a_1(xc) + \cdots + a_m(x^m c).$$

Each term lies in $\Phi^{-1}(C)$, so $a(x)c$ also lies in $\Phi^{-1}(C)$. Thus $\Phi^{-1}(C)$ is an ideal of A.

Conversely, let $J \triangleleft A$ be an ideal. Then J is in particular an $\mathbb{F}_q$-subspace of A, and since J is stable under multiplication by x, its image $\Phi(J) \subseteq \mathbb{F}_q^n$ is stable under cyclic shift. Hence $\Phi(J)$ is a cyclic code. ☐

Combining this theorem with Proposition 8.1, we obtain the classification of cyclic codes.

Corollary 10.3 (Classification of cyclic codes)**.** *Every cyclic code of length n over $\mathbb{F}_q$ is of the form*

$$C = \langle \overline{g(x)} \rangle \subseteq \mathbb{F}_q[x]/\langle x^n - 1\rangle,$$

for a unique monic divisor $g(x)$ of $x^n - 1$.

The polynomial $g(x)$ is called the *generator polynomial* of the cyclic code.

11 Dimension and basis of a cyclic code

Let

$$C = \langle \overline{g(x)} \rangle \subseteq \mathbb{F}_q[x]/\langle x^n - 1 \rangle,$$

and write

$$x^n - 1 = g(x)h(x).$$

Set

$$r = \deg g.$$

Theorem 11.1. *The cyclic code C has dimension*

$$\dim_{\mathbb{F}_q} C = n - r.$$

Moreover, the vectors corresponding to

$$\overline{g(x)},\ \overline{xg(x)},\ \overline{x^2 g(x)},\ \ldots,\ \overline{x^{n-r-1} g(x)}$$

form a basis of C.

Proof. Define an $\mathbb{F}_q$-linear map

$$\psi : \{m(x) \in \mathbb{F}_q[x] : \deg m < n - r\} \longrightarrow C, \qquad m(x) \longmapsto \overline{m(x)g(x)}.$$

We first show that ψ is injective. Suppose

$$\overline{m(x)g(x)} = 0$$

in A. Then

$$x^n - 1 = g(x)h(x) \mid m(x)g(x)$$

in $\mathbb{F}_q[x]$. Since $\mathbb{F}_q[x]$ is an integral domain, this implies

$$h(x) \mid m(x).$$

But

$$\deg m < n - r = \deg h,$$

so $m(x) = 0$.

Next we show that ψ is surjective. Any element of C has the form

$$\overline{a(x)g(x)}$$

for some $a(x) \in \mathbb{F}_q[x]$. Divide $a(x)$ by $h(x)$:

$$a(x) = q(x)h(x) + r(x), \qquad \deg r < \deg h = n - r.$$

Then

$$a(x)g(x) = q(x)h(x)g(x) + r(x)g(x) = q(x)(x^n - 1) + r(x)g(x),$$

so in the quotient ring,

$$\overline{a(x)g(x)} = \overline{r(x)g(x)}.$$

Hence every codeword lies in the image of ψ.

Therefore ψ is an isomorphism onto C. Its domain has dimension $n - r$, with basis

$$1, x, \ldots, x^{n-r-1}.$$

Transporting this basis through ψ gives the desired basis of C. $\square$

12 Parity-check polynomial

Let

$$C = \langle \overline{g(x)} \rangle \subseteq A, \qquad x^n - 1 = g(x)h(x).$$

The polynomial $h(x)$ is called the *parity-check polynomial* of the cyclic code.

Theorem 12.1 (Kernel description)**.** *Let*

$$\mu_h : A \longrightarrow A, \qquad f \longmapsto fh.$$

Then

$$C = \mathrm{Ker}(\mu_h).$$

Equivalently,

$$C = \{\, c(x) \in A : c(x)h(x) = 0 \,\}.$$

Proof. If $c(x) \in C$, then

$$c(x) = m(x)g(x)$$

for some $m(x)$. Hence

$$c(x)h(x) = m(x)g(x)h(x) = m(x)(x^n - 1) = 0$$

in A. Thus $C \subseteq \mathrm{Ker}(\mu_h)$.

Conversely, suppose $c(x)h(x) = 0$ in A. Then

$$x^n - 1 = g(x)h(x) \mid c(x)h(x)$$

in $\mathbb{F}_q[x]$. Since $\mathbb{F}_q[x]$ is an integral domain and $h(x) \neq 0$, we conclude that

$$g(x) \mid c(x).$$

Hence $c(x) \in \langle \overline{g(x)} \rangle = C$. $\square$

This theorem explains the name *parity-check polynomial*: multiplication by $h(x)$ gives a linear condition that exactly characterizes the code.

Remark 12.2. There is a perfect formal symmetry:

$$C = \mathrm{Im}(\mu_g) = \mathrm{Ker}(\mu_h).$$

Thus the generator polynomial $g(x)$ describes the code as an image, while the parity-check polynomial $h(x)$ describes the code as a kernel.

13 The spectrum of the cyclic-code ring

Geometrically, the quotient map

$$\mathbb{F}_q[x] \longrightarrow \mathbb{F}_q[x]/\langle x^n - 1 \rangle$$

induces a closed immersion

$$\mathrm{Spec}(\mathbb{F}_q[x]/\langle x^n - 1 \rangle) \hookrightarrow \mathrm{Spec}(\mathbb{F}_q[x]).$$

Its image is

$$V(x^n - 1).$$

If

$$x^n - 1 = \prod_{i=1}^{r} f_i(x)^{e_i}$$

is the factorization into monic irreducibles, then the points of this spectrum correspond to the prime ideals

$$(f_1), \ldots, (f_r).$$

Remark 13.1. Topologically, only the distinct irreducible factors matter. Scheme-theoretically, the exponents e_i are still present through nilpotent structure.

If $x^n - 1$ is squarefree, then the Chinese remainder theorem yields

$$\mathbb{F}_q[x]/\langle x^n - 1\rangle \cong \prod_{i=1}^{r} \mathbb{F}_q[x]/\langle f_i(x)\rangle.$$

Hence

$$\operatorname{Spec}(\mathbb{F}_q[x]/\langle x^n - 1\rangle) \cong \coprod_{i=1}^{r} \operatorname{Spec}(\mathbb{F}_q[x]/\langle f_i(x)\rangle)\,.$$

This is the finite-field analogue of the squarefree decomposition for $\mathbb{C}[x]/\langle g(x)\rangle$.

14 A small worked example

Take $q = 2$ and $n = 3$. Then

$$x^3 - 1 = x^3 + 1 = (x + 1)(x^2 + x + 1)$$

in $\mathbb{F}_2[x]$.

The monic divisors of $x^3 - 1$ are

$$1, \qquad x + 1, \qquad x^2 + x + 1, \qquad x^3 + 1.$$

Hence there are exactly four cyclic codes of length 3 over $\mathbb{F}_2$.

(1) $g(x) = 1$. Then

$$C = \langle \overline{1}\rangle = \mathbb{F}_2[x]/\langle x^3 - 1\rangle \cong \mathbb{F}_2^3.$$

(2) $g(x) = x + 1$. Then $\deg g = 1$, so C has dimension 2. Its basis is

$$\overline{x + 1}, \qquad \overline{x(x + 1)} = \overline{x^2 + x}.$$

As vectors, these are

$$(1, 1, 0), \qquad (0, 1, 1).$$

Thus the code is

$$\{000,\ 011,\ 101,\ 110\},$$

the even-weight code.

(3) $g(x) = x^2 + x + 1$. Then $\deg g = 2$, so C has dimension 1. Its basis vector is

$$\overline{x^2 + x + 1} \leftrightarrow (1, 1, 1).$$

Thus the code is

$$\{000,\ 111\},$$

the repetition code.

(4) $g(x) = x^3 + 1 = x^3 - 1$. Then

$$C = \langle \overline{x^3 - 1} \rangle = 0.$$

This example is the exact analogue of the ideals of $\mathbb{Z}/12\mathbb{Z}$: divisors upstairs classify ideals downstairs.

15 A final dictionary

We close with a compact dictionary that summarizes the whole discussion.

Algebra

Ring	Quotient	Ideals downstairs
$\mathbb{Z}$	$\mathbb{Z}/(12)$	divisors of 12
$\mathbb{C}[x]$	$\mathbb{C}[x]/\langle g(x) \rangle$	divisors of $g(x)$
$\mathbb{F}_q[x]$	$\mathbb{F}_q[x]/\langle x^n - 1 \rangle$	divisors of $x^n - 1$

Geometry

$$\operatorname{Spec}(R/I) \cong V(I) \subseteq \operatorname{Spec}(R).$$

For $R = \mathbb{Z}$, quotienting by (12) cuts out the points (2) and (3).

For $R = \mathbb{C}[x]$, quotienting by (g) cuts out the roots of g.

For $R = \mathbb{F}_q[x]$, quotienting by $(x^n - 1)$ cuts out the closed subscheme $V(x^n - 1)$.

Coding theory

$$A = \mathbb{F}_q[x]/\langle x^n - 1 \rangle \cong \mathbb{F}_q^n.$$

Multiplication by x corresponds to cyclic shift. Therefore

$$\text{ideals of } A \quad \Longleftrightarrow \quad \text{cyclic codes of length } n.$$

If

$$x^n - 1 = g(x)h(x),$$

then the cyclic code generated by $g(x)$ satisfies

$$C = \langle \overline{g(x)} \rangle = \operatorname{Im}(\mu_g) = \operatorname{Ker}(\mu_h).$$

$$\boxed{\text{cyclic code} \ = \ \text{ideal of } \mathbb{F}_q[x]/\langle x^n - 1 \rangle \ = \ \text{subspace of } \mathbb{F}_q^n \text{ stable under cyclic shift.}}$$